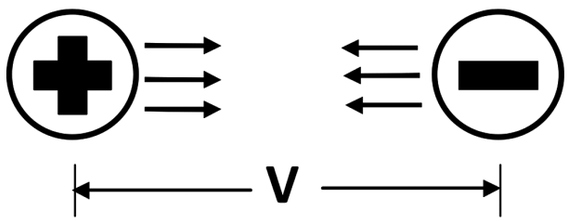
10.2 BASICS OF ELECTRICITY?

Electricity is one type of energy. All matters whether solid, liquid or gaseous consist of minute particles known as Atom. According to modern research electric current means electrons movement only.

Voltage

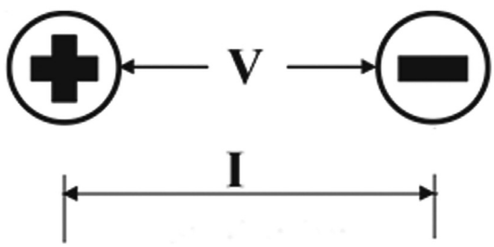
The electric pressure which is used to move electrons from one end to another end is called voltage. It is represented by the letter 'V' and the unit is volts. It is measured by voltmeter. Similarities to voltage are volt, emf, potential and potential difference.



Voltage

Current

Flow of Electron in a conductor is called current. It is represented by the letter 'I' and the unit is called ampere(A). Current can be measured by ammeter.



Current

Resistance

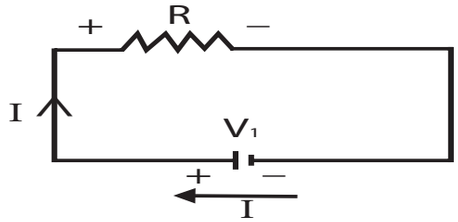
Resistance may be defined as the property of a substance which opposes the flow of current flowing through it. It is represented by the letter R and the unit is ohm(Ω). It is measured by ohm meter. Mega ohms value is measured by using meggar

Example: when compare gold to iron, gold has less resistance value. Resistance depends upon materials.

Also the resistance depends on the temperature of co-efficient of the material.

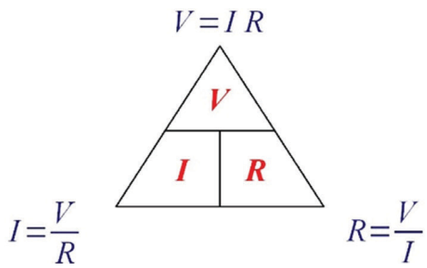
OHM'S LAW

Ohm's law states that, "In any closed circuit the current is directly proportional to the voltage applied and inversely proportional to the resistance of the circuit at a constant temperature".



i.e: I α V and I α 1/R from this

V= I × R and R=V/I



Where,

I = current,

V = voltage,

R = resistance.

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